Fiscal Year:	FY 2025	**Task Last Updated:**	FY 09/02/2024
PI Name:	Weaver, Ashley Ph.D.		
Project Title:	Quantitative CT and MRI-based Modeling Assessment of Dynamic Vertebral Strength and Injury Risk Following Long-Duration Spaceflight		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Space Human Factors Engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **Dynamic Loads**:Risk of In-Mission Injury and Performance Decrements and Long-term Health Effects due to Dynamic Loads		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	UNIVERSITY	**Phone:**	336-716-0944
Organization Name:	Wake Forest University		
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City:	Winston-Salem	**State:**	NC
Zip Code:	27101-4101	**Congressional District:**	12
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2015-16 HERO NNJ15ZSA001N-Crew Health (FLAGSHIP, NSBRI, OMNIBUS). Appendix A-Crew Health, Appendix B-NSBRI, Appendix C-Omnibus
Start Date:	10/01/2016	**End Date:**	08/31/2025
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
Contact Email:	alexandra.m.whitmire@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 8/31/2025 per NSSC information (Ed., 7/12/24) NOTE: End date changed to 8/31/2024 per S. Mack-Phillips/JSC (Ed., 7/18/23) NOTE: End date changed to 8/31/2023 per V. Lehman/JSC (Ed., 3/29/23) NOTE: End date changed to 8/31/2022 per L. Barnes-Moten/JSC (Ed., 8/2/21) NOTE: End date changed to 8/31/2021 per NSSC information/S. Huppman/HRP (Ed., 2/25/2020) NOTE: End date changed to 2/1/2020 per NSSC information (Ed., 7/8/19)		
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):	Stitzel, Joel Ph.D. (Wake Forest University) Tooze, Janet Ph.D. (Wake Forest University)		

Grant/Contract No.:	NNX16AP89G
Performance Goal No.:	
Performance Goal Text:	
Task Description:	Prolonged periods of near weightlessness can cause damage to astronauts' musculoskeletal system. This damage can increase the risk of skeletal tissue failure (e.g., fractures, tears) when experiencing forceful, dynamic loads. Fractures of the spine during dynamic conditions such as launch or landing could cause a mission to fail. This study will measure this degradation of astronauts' vertebrae and spinal muscles during missions aboard the International Space Station (ISS). We will then determine the extent of vertebral weakening of crewmembers during long-duration missions. Changes in pre- and post-flight vertebral geometry, volume, cortex thickness, and bone mineral density will be measured from existing lumbar quantitative computed tomography (qCT) scans, as well as from planned qCT scans of the cervical, thoracic, and lumbar spine from nine ISS crewmembers. Likewise, the pre- and post-flight spinal muscle volumes will be analyzed using both existing magnetic resonance imaging (MRI) scans and planned MRI scans from nine ISS crewmembers. The qCT and MRI scans will be analyzed to determine structural and material changes in the cervical, thoracic, and lumbar vertebrae and the spinal muscles that indicate damage which could weaken these tissues. Our unique engineering approach will measure the loss of vertebral strength during spaceflight conditions and predict the risk of failure during traumatic, dynamic loading conditions such as launch or landing. Vertebral strength and risk for vertebral fracture and injury will be quantified in dynamic simulations using a full human body model that is constructed using structural and material data gathered from the pre- and post-flight medical images for each astronaut. This study has significance in quantifying and addressing risks of long-duration spaceflight, including vertebral injury from dynamic loads, vertebral fracture, early onset vertebral osteoporosis due to spaceflight, and impaired performance due to reduced spinal muscle mass, strength, and endurance.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Microgravity induces similar spinal changes to those seen in the aging population and people with limited mobility. Demonstrating how the vertebral column changes in response to microgravity can aid in refining the diagnostic and treatment protocols of physicians on Earth. Additionally, assessing vertebral column strength using finite element modeling can provide future techniques for assessing the efficacy of osteoporosis treatments, which would particularly benefit older adults.
Task Progress:	Prospective quantitative computed tomography (qCT) and magnetic resonance imaging (MRI) data collection has been completed for all nine of the enrolled subjects. Bone and muscle morphology and quality measures were completed. Vertebral compression test simulations were conducted to assess effects of space-induced changes on vertebral compression strength using subject-specific vertebrae models. To study the effects of musculoskeletal changes on astronaut spinal injury risk, a modular deformable spine finite element human body model was developed by incorporating a deformable spine in the existing simplified human body model. Morphing has been used to create subject-specific versions of this model that match data derived from the astronaut's imaging. Work under student augmentation awards including human body modeling of lunar launch/landing events with small, medium, and large sizes of a human body model.
Bibliography Type:	Description: (Last Updated: 01/20/2026)
Abstracts for Journals and Proceedings	Weaver AA, Jeong T, Fuentes DA, Poveda L, Blade S, Mumtaz M, Devane K. "Spaceflight effects on spine musculoskeletal health in astronauts: A QCT, MRI, and finite element analysis study." 24th International Workshop on Quantitative Musculoskeletal Imaging (QMSKI) Conference, Adelaide, Australia, November 3-8, 2024. Abstracts. 24th International Workshop on Quantitative Musculoskeletal Imaging (QMSKI) Conference, Adelaide, Australia, November 3-8, 2024. , Nov-2024
Abstracts for Journals and Proceedings	Thrush K, Mumtaz M, Jeong T, Poveda L, Madrid D, Devane K, Weaver AA. "Impact of microgravity on astronauts' vertebral bone and muscle health." Biomedical Engineering Society (BMES) 2024 Annual Meeting, Baltimore, MD, October 23-26, 2024. Abstracts. Biomedical Engineering Society (BMES) 2024 Annual Meeting, Baltimore, MD, October 23-26, 2024. , Oct-2024